Electronics and
Computer Science

UNIVERSITY OF
Southampton

AICE2008 Computer Architecture

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1. Introduction

What is this module about?

- How do processors (CPUs) work?
- How to implement a CPU?
- What is an Instruction Set Architecture?
- How to improve performance?
- How to incorporate a CPU in a larger system?

- Much of this was covered in AICE1002 last year, but we're going to go into more depth

Why is this important?

- You will probably never design a new processor
- You might work for a company that designs CPUs, e.g. Arm
- You might build systems that include CPUs
- You might design accelerators for CPUs, e.g. for AI/ML
- If you understand how a CPU works, you will probably write better software

Outline Syllabus

- Classic architectures:
  - ISAs
  - Data-path and control-path design
  - Detailed single-cycle CPU for real-world ISA
- Optimisations:
  - Pipelining
  - Caches
  - Memory hierarchy
- Bus-based systems:
  - Buses & protocols
  - Polling/interrupts
  - IP interfacing

Warning!
This is the first year that we have run this module.
Details may change.

Assessment

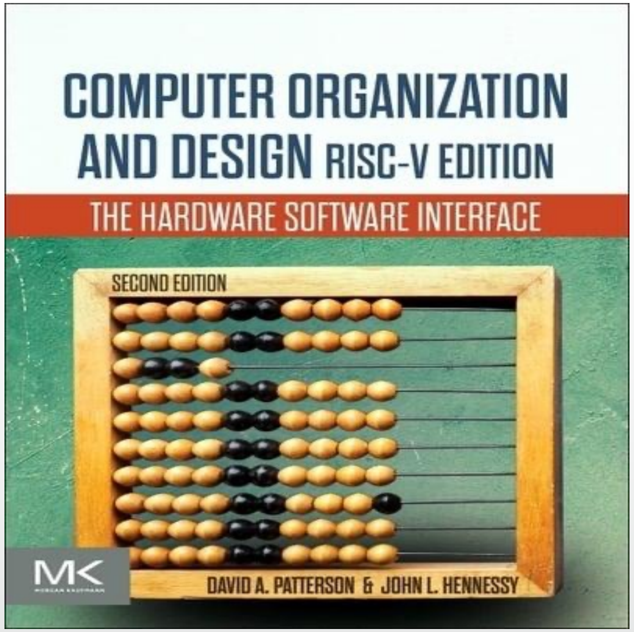
- Exam – 40%
- Labs, design exercise – 60%
  - Schedule to be confirmed
  - Probably:
    - Introductory lab
    - 2 labs before Easter
    - Design exercise after Easter

Resources

- Moodle
  - Lecture slides
  - Recordings
- FPGA development boards
- Software on your laptops

Book

- "Computer Organization and Design RISC-V Edition The Hardware Software Interface" David A. Patterson, John L. Hennessy
  - Available as eBook from University Library



Note. This module is not just about RISC-V, but we will use RISC-V for labs and examples.

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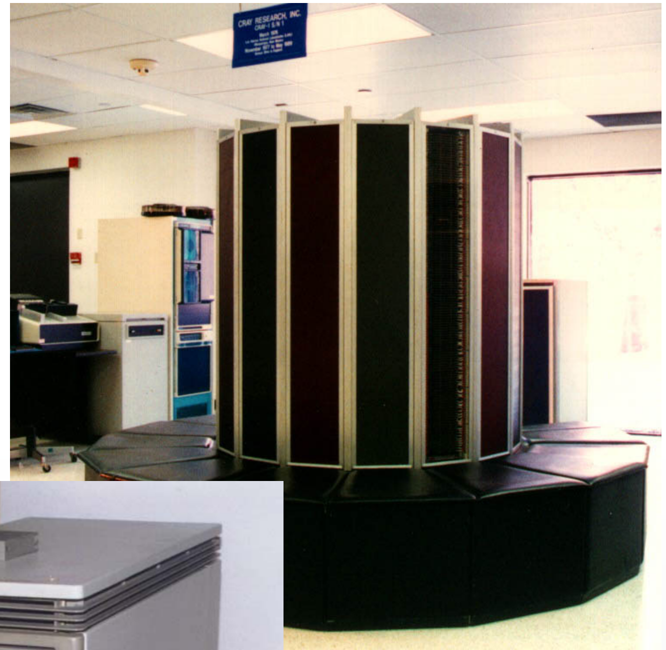
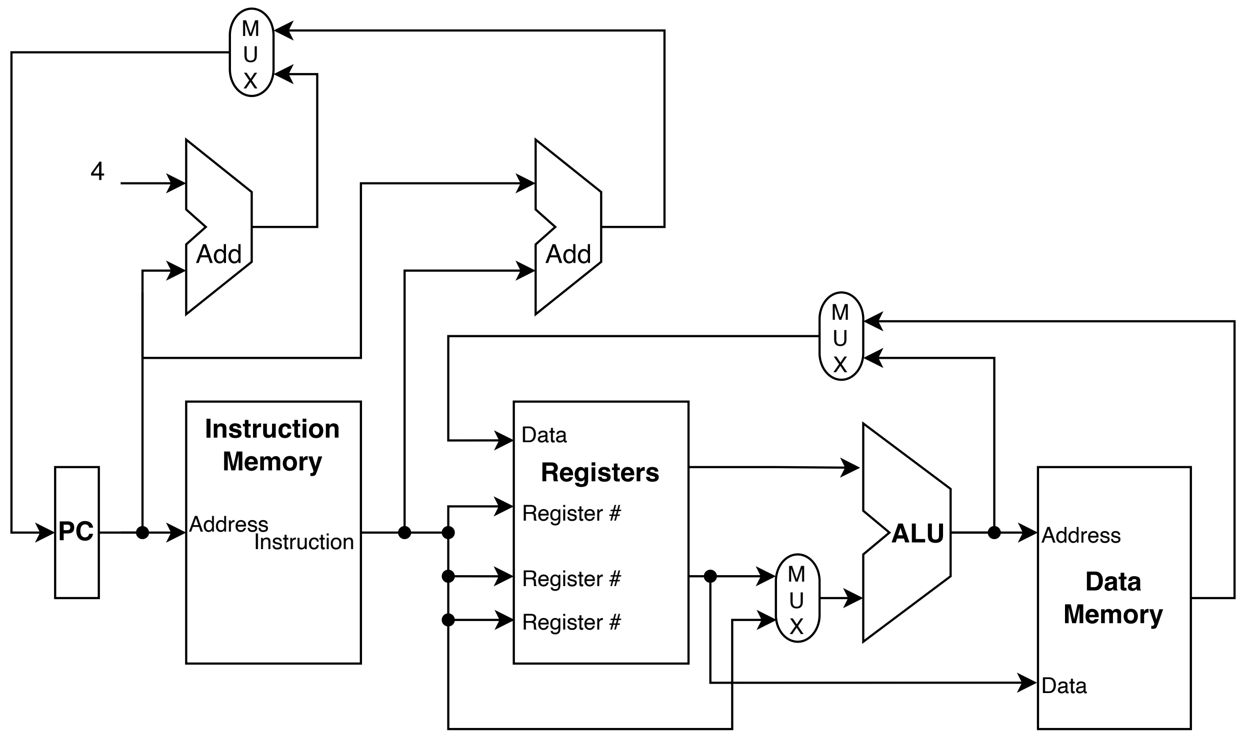
Some Ancient History



Different designs;
Different objectives;
Many of the same ideas
still apply!

RISC-V



- Doesn't show control signals
- Single cycle
- Basic functional blocks

Instructions per Cycle

- An important measure of performance is Instructions per Cycle (IPC)
- or the reciprocal, Cycles per Instruction (CPI)
- For a single cycle processor, IPC = 1
  - This is good, but the clock speed may be low
  - We'll look at this next